Physica bk7

May 17, 2026

1 On GlossySnake

A LaTeX Proof-of-Concept of the GlossySnake project. By the nature of the machine-glossing, the inaccurate and malformatted glossing is expected.

2 Physica bk7

PHYSIKA Bk.7

Physics Book 7

Κεφάλαιο 1

Chapter 1

Ἅπαν τὸ κινούμενον ὑπό τινος ἀνάγκη κινεῖσθαι· εἰ μὲν γὰρ ἐν

All the moving by someone necessity to be moved. If indeed for in

ἑαυτῷ μὴ ἔχει τὴν ἀρχὴν τῆς κινήσεως, φανερὸν ὅτι ὑφ’

itself not has the origin of the motion, it is clear that it is moved by

ἑτέρου κινεῖται (ἄλλο γὰρ ἔσται τὸ κινοῦν)· εἰ δ’ ἐν αὑτῷ,

another (it moves) (another for it will be the mover). If but in itself,

ἔστω [τὸ] εἰλημμένον ἐφ’ οὗ τὸ AB ὃ κινεῖται καθ’

let it be [the] taken upon which the AB which is moved according to

αὑτό, ἀλλὰ μὴ <τῷ τῶν> τούτου τι κινεῖσθαι. πρῶτον

itself, but do not by the of the this anything to be moved. First

μὲν οὖν τὸ ὑπολαμβάνειν τὸ AB ὑφ’ ἑαυτοῦ κινεῖσθαι διὰ

indeed therefore the assumption the AB by itself to be moved through

τὸ ὅλον τε κινεῖσθαι καὶ ὑπ' οὐδενὸς τῶν ἔξωθεν ὅμοιόν ἐστιν

the whole and to be moved and by no of the external similar it is

ὥσπερ εἰ τοῦ ΚΛ κινοῦντος τὸ ΛΜ καὶ αὐτοῦ κινουμένου εἰ μὴ

as if of the KL moving the LM and itself of the moving if not

φάσκοι τις τὸ ΚΜ κινεῖσθαι ὑπό τινος, διὰ τὸ

one should say anyone the K.M. (abbreviation) to be moved by someone, because the

μὴ φανερὸν εἶναι πότερον τὸ κινοῦν καὶ πότερον τὸ κινούμενον·

not evident to be whether the moving and whether the moved.

εἶτα τὸ μὴ ὑπό τινος κινούμενον [242a] οὐκ ἀνάγκη παύσασθαι

Then the not by someone moving [242a] not necessity to stop

κινούμενον τῷ ἄλλο ἠρεμεῖν, ἀλλ' εἴ τι ἠρεμεῖ τῷ ἄλλο

moving to the other to be at rest, but if anything is at rest to the other

πεπαῦσθαι κινούμενον, ἀνάγκη ὑπό τινος αὐτὸ κινεῖσθαι. τούτου

having ceased moving, necessity by someone it must be moved. Of this

δ' εἰλημμένου πᾶν τὸ κινούμενον κινήσεται ὑπό τινος. ἐπεὶ γὰρ

but having been taken all the moving will be moved by someone. Since for

εἴληπται [τὸ] κινούμενον ἐφ' ᾧ τὸ ΑΒ, ἀνάγκη διαιρετὸν

it has been taken [the] moving on which the A.B., necessity to be divisible

αὐτὸ εἶναι· πᾶν γὰρ τὸ κινούμενον διαιρετόν. διῃρήσθω δὴ

it to be. All for the moving is divisible. Let it be divided indeed

κατὰ τὸ Γ. τοῦ δὴ ΓΒ μὴ κινουμένου οὐ κινηθήσεται τὸ

according to the G. of the indeed GB not moving not will be moved the

ΑΒ. εἰ γὰρ κινήσεται, δῆλον ὅτι τὸ ΑΓ κινοῖτ' ἂν τοῦ ΓΒ

AB. if for it will move, it is clear that the AG moves if of the GB

ἠρεμοῦντος, ὥστε οὐ καθ' αὑτὸ κινηθήσεται καὶ πρῶτον. ἀλλ'

at rest, so that not according to itself will be moved and first. but

ὑπέκειτο καθ' αὑτὸ κινεῖσθαι καὶ πρῶτον. ἀνάγκη ἄρα

it was underlying according to itself to be moved and first. necessity therefore

τοῦ ΓΒ μὴ κινουμένου ἠρεμεῖν τὸ ΑΒ. ὃ δὲ ἠρεμεῖ (45)

of the GB not moving to be at rest the AB. which but is at rest (45)

μὴ κινουμένου τινός, ὡμολόγηται ὑπό τινος κινεῖσθαι, ὥστε

not moving of anything, it has been agreed by someone to be moved, so that

πᾶν ἀνάγκη τὸ κινούμενον ὑπό τινος κινεῖσθαι· ἀεὶ γὰρ

everything necessity the moving by something to be moved· always for

ἔσται τὸ κινούμενον διαιρετόν, τοῦ δὲ μέρους μὴ κινουμένου

it will be the moving divisible, of the but part not moving

ἀνάγκη καὶ τὸ ὅλον ἠρεμεῖν.

necessity and the whole to be at rest.

ἐπεὶ δὲ πᾶν τὸ κινούμενον ἀνάγκη κινεῖσθαι ὑπό τινος, ἐάν γέ

since but all the moving necessity to be moved by someone, if at least

τι κινῆται τὴν ἐν τόπῳ κίνησιν ὑπ' ἄλλου κινουμένου, καὶ

something moves the in place motion by another moving, and

πάλιν τὸ κινοῦν ὑπ' ἄλλου κινουμένου κινῆται κἀκεῖνο ὑφ' ἑτέρου

again the mover by another moving is moved and that by another

καὶ ἀεὶ οὕτως, ἀνάγκη εἶναί τι τὸ πρῶτον κινοῦν, καὶ μὴ

and always thus, necessity to be something the first mover, and not

βαδίζειν εἰς ἄπειρον· μὴ γὰρ ἔστω, ἀλλὰ γενέσθω ἄπειρον.

to walk into the infinite. not for let there be, but let it become infinite.

κινείσθω δὴ τὸ μὲν Α ὑπὸ τοῦ Β, τὸ δὲ Β ὑπὸ τοῦ Γ,

let it be moved indeed the one A by the B, the other B by the C,

τὸ δὲ Γ ὑπὸ τοῦ Δ, καὶ ἀεὶ τὸ ἐχόμενον ὑπὸ τοῦ ἐχομένου.

the other C by the D, and always the held by the holding.

ἐπεὶ οὖν ὑπόκειται τὸ κινοῦν κινούμενον κινεῖν, ἀνάγκη ἅμα

since therefore is underlying the mover being moved to move, necessity together

γίγνεσθαι τὴν τοῦ κινουμένου καὶ τὴν τοῦ κινοῦντος κίνησιν
to become the of the being moved and the of the mover motion
(ἅμα γὰρ κινεῖ τὸ κινοῦν καὶ κινεῖται τὸ κινούμενον)·
(together for moves the mover and is being moved the being moved)
φανερὸν <οὖν> ὅτι ἅμα ἔσται τοῦ A καὶ τοῦ B καὶ τοῦ Γ
evident <therefore> that together will be of the A and of the B and of the C
καὶ ἑκάστου τῶν κινούντων καὶ κινουμένων ἡ κίνησις. εἰλήφθω
and each of the movers and being moved the motion. Let it be taken
οὖν ἡ ἑκάστου κίνησις, καὶ ἔστω τοῦ μὲν A ἐφ' ἧς
therefore the each motion, and let it be of the on the one hand A on which
E, τοῦ δὲ B ἐφ' ἧς Z, τῶν <δὲ> Γ Δ ἐφ' ὧν H
E, of the on the other hand B on which Z, of the <but> C D on which H
Θ. εἰ γὰρ ἀεὶ κινεῖται ἕκαστον ὑφ' ἑκάστου, ὅμως ἔσται λαβεῖν
θ. If for always it is moved each under each, however will be to take
μίαν ἑκάστου κίνησιν τῷ ἀριθμῷ· πᾶσα γὰρ κίνησις ἔκ τινος
one of each movement to the number. every for movement from someone
εἴς τι, καὶ οὐκ ἄπειρος τοῖς ἐσχάτοις· λέγω δὴ ἀριθμῷ
into something, and not infinite to the last. I say indeed to the number
μίαν κίνησιν τὴν ἐκ τοῦ αὐτοῦ εἰς τὸ αὐτὸ τῷ ἀριθμῷ ἐν τῷ
one movement the from of the same into the same to the number in the
αὐτῷ χρόνῳ τῷ ἀριθμῷ γιγνομένην. ἔστι γὰρ κίνησις καὶ γένει
same time the number being generated. there is for movement and by kind
καὶ εἴδει καὶ ἀριθμῷ ἡ αὐτή, [242b] γένει μὲν ἡ τῆς αὐτῆς
and by type and by number the same, [242b] by kind on one hand the of the same
κατηγορίας, οἷον οὐσίας ἢ ποιότητος, εἴδει δὲ <ἡ> ἐκ τοῦ
category, such as essence or quality, by type but <the> from of the
αὐτοῦ τῷ εἴδει εἰς τὸ αὐτὸ τῷ εἴδει, οἷον ἐκ λευκοῦ εἰς
same to the type into the same to the type, such as from white into

μέλαν ἢ ἐξ ἀγαθοῦ εἰς κακὸν ἀδιάφορον τῷ εἴδει· ἀριθμῷ δὲ ἡ

black or from good into bad indifferent the kind; number but the

ἐξ ἑνὸς τῷ ἀριθμῷ <εἰς ἓν τῷ ἀριθμῷ> ἐν τῷ αὐτῷ χρόνῳ, οἷον

from one the number into one the number> in the same time, such as

ἐκ τοῦδε τοῦ λευκοῦ εἰς τόδε τὸ μέλαν, ἢ ἐκ τοῦδε τοῦ τόπου

from this the white into this the black, or from this the place

εἰς τόνδε, ἐν τῷδε τῷ χρόνῳ· εἰ γὰρ ἐν ἄλλῳ, οὐκέτι ἔσται

into this, in this the time; if for in another, no longer will be

ἀριθμῷ μία κίνησις, ἀλλ' εἴδει.

number one motion, but kind.

εἴρηται δὲ περὶ τούτων ἐν τοῖς πρότερον. εἰλήφθω

it has been stated but concerning these in the earlier. let it be taken

δὲ καὶ ὁ χρόνος ἐν ᾧ κεκίνηται τὴν αὐτοῦ κίνησιν τὸ Α, καὶ

but also the time in which is moved the its motion the A, and

ἔστω ἐφ' ᾧ Κ. πεπερασμένης δ' οὔσης τῆς τοῦ Α κινήσεως

let it be on which K. being finite but being the of the A motion

καὶ ὁ χρόνος ἔσται πεπερασμένος. ἐπεὶ δὴ ἄπειρα τὰ κινοῦντα

and the time will be finite. since indeed infinite the movers

καὶ τὰ κινούμενα, καὶ ἡ κίνησις ἡ ΕΖΗΘ ἡ ἐξ ἁπασῶν ἄπειρος

and the moved, and the motion the EZHΘ the from all infinite

ἔσται· ἐνδέχεται μὲν γὰρ ἴσην εἶναι τὴν τοῦ Α καὶ τοῦ

will be. it may happen on the one hand for equal to be the of the A and of the

Β καὶ τὴν τῶν ἄλλων, ἐνδέχεται δὲ μείζους τὰς τῶν ἄλλων, ὥστε

B and the of the others, it may happen but greater the of the others, so that

εἴ τε ἴσαι εἴ τε μείζους, ἀμφοτέρως ἄπειρος ἡ ὅλη· λαμβάνομεν

if and equal if and greater, both infinite the whole· we take

γὰρ τὸ ἐνδεχόμενον. ἐπεὶ δ' ἅμα κινεῖται καὶ τὸ Α καὶ τῶν

for the possible. since but together moves and the A and of the

ἄλλων ἕκαστον, ἡ ὅλη κίνησις ἐν τῷ αὐτῷ χρόνῳ ἔσται καὶ ἡ
others each, the whole movement in the same time will be and the
τοῦ A· ἡ δὲ τοῦ A ἐν πεπερασμένῳ· ὥστε εἴη ἂν
of the A· the but of the A in finite· so that it would be could
ἄπειρος ἐν πεπερασμένῳ, τοῦτο δ' ἀδύνατον.
infinite in finite, this but impossible.

οὕτω μὲν οὖν δόξειεν ἂν δεδεῖχθαι τὸ ἐξ
thus indeed therefore it would seem (particle) to have been demonstrated the from
ἀρχῆς, οὐ μὴν ἀποδείκνυται διὰ τὸ μηδὲν δείκνυσθαι
the beginning, not nevertheless it is shown through the nothing to be demonstrated
ἀδύνατον· ἐνδέχεται γὰρ ἐν πεπερασμένῳ χρόνῳ ἄπειρον εἶναι
is impossible. it may happen for in the finite time is infinite to be
κίνησιν, μὴ ἑνὸς ἀλλὰ πολλῶν. ὅπερ συμβαίνει καὶ ἐπὶ τούτων·
movement, not of one but of many. which happens and also in these;
ἕκαστον γὰρ κινεῖται τὴν ἑαυτοῦ κίνησιν, ἅμα δὲ πολλὰ
each for is moved its own movement, at the same time but many
κινεῖσθαι οὐκ ἀδύνατον. ἀλλ' εἰ τὸ κινοῦν πρώτως κατὰ τόπον
to be moved is not impossible. but if the mover first in terms of place
καὶ σωματικὴν κίνησιν ἀνάγκη ἢ ἅπτεσθαι ἢ συνεχὲς εἶναι τῷ
and bodily movement necessity or to touch or continuously to be with the
κινουμένῳ, καθάπερ ὁρῶμεν ἐπὶ πάντων, ἀνάγκη τὰ κινούμενα
moving as we see we observe upon all, necessity the moved
καὶ τὰ κινοῦντα συνεχῆ εἶναι ἢ ἅπτεσθαι ἀλλήλων, ὥστ'
and the movers to be continuous to be or to touch of each other, so that
εἶναί τι ἐξ ἁπάντων ἕν. τοῦτο δὲ εἴτε πεπερασμένον εἴτε
to be what from all one. this but whether finite or

ἄπειρον, οὐδὲν διαφέρει πρὸς τὰ νῦν· πάντως γὰρ ἡ κίνησις

infinite, nothing differs in the now· certainly for the motion

ἔσται ἄπειρος ἀπείρων ὄντων, εἴπερ ἐνδέχεται καὶ ἴσας εἶναι

will be infinite of infinitely existing, if it is possible and equal to be

καὶ μείζους ἀλλήλων· ὃ γὰρ ἐνδέχεται, ληψόμεθα ὡς ὑπάρχον.

and greater than each other· what for it is possible, we will take as existing.

εἰ οὖν τὸ μὲν ἐκ τῶν Α Β Γ Δ <ἢ πεπερασμένον ἢ>

if therefore the on the one hand from the A B C D or < finite or>

ἄπειρόν τί ἐστιν, κινεῖται δὲ τὴν ΕΖΗΘ κίνησιν ἐν τῷ χρόνῳ

infinite what is, moves but the EZHΘ motion in the time

τῷ Κ, οὗτος δὲ πεπέρανται, συμβαίνει ἐν πεπερασμένῳ χρόνῳ

the K, this but extends, it happens in finite time

ἄπειρον διιέναι ἢ τὸ πεπερασμένον ἢ τὸ ἄπειρον. ἀμφοτέρως δὲ

infinite to go through or the finite or the infinite. in both ways but

ἀδύνατον· ὥστε ἀνάγκη ἵστασθαι καὶ εἶναί τι πρῶτον κινοῦν

impossible; so that necessity to stand and to be something first causing

καὶ κινούμενον. οὐδὲν γὰρ διαφέρει τὸ συμβαίνειν ἐξ ὑποθέσεως

and being moved. nothing for differs the occurring from hypothesis

τὸ ἀδύνατον· [243a] ἡ γὰρ ὑπόθεσις εἴληπται ἐνδεχομένη, τοῦ

the impossible; [243a] the for hypothesis has been taken as possible, of the

δ' ἐνδεχομένου τεθέντος οὐδὲν προσήκει γίγνεσθαι διὰ

of the possible having been set nothing is suitable to happen because of

τοῦτο ἀδύνατον.

this impossible.

Κεφάλαιο 2

Chapter 2

Τὸ δὲ πρῶτον κινοῦν, μὴ ὡς τὸ οὗ ἕνεκεν, ἀλλ' ὅθεν ἡ

The but first moving, not as the for which sake, but from where the

ἀρχὴ τῆς κινήσεως, ἅμα τῷ κινουμένῳ ἐστί (λέγω δὲ τὸ

beginning of the motion, together with the moving is (I say but the

ἅμα, ὅτι οὐδέν ἐστιν αὐτῶν μεταξύ)· τοῦτο γὰρ κοινὸν ἐπὶ

together, that nothing exists of them in between)· this for common to

παντὸς κινουμένου καὶ κινοῦντός ἐστιν. ἐπεὶ δὲ τρεῖς αἱ κινήσεις,

all moving and moving is. Since but three the motions,

ἥ τε κατὰ τόπον καὶ ἡ κατὰ τὸ ποιὸν καὶ ἡ κατὰ

which also according to place and the according to the kind and the according to

τὸ ποσόν, ἀνάγκη καὶ τὰ κινοῦντα τρία εἶναι, τό τε φέρον καὶ τὸ

the quantity, necessity and the movers three to be, the and carrying and the

ἀλλοιοῦν καὶ τὸ αὔξον ἢ φθῖνον.

altering and the increasing or diminishing.

πρῶτον οὖν εἴπωμεν περὶ τῆς φορᾶς· πρώτη γὰρ αὕτη τῶν

first therefore let us say about the motion: first for this of the

κινήσεων. ἅπαν δὴ τὸ φερόμενον ἢ ὑφ' ἑαυτοῦ κινεῖται ἢ

movements. all indeed the carried or by itself is moved or

ὑπ' ἄλλου. ὅσα μὲν οὖν αὐτὰ ὑφ' αὑτῶν κινεῖται,

by another. as much on the one hand therefore they by themselves are moved,

φανερὸν ἐν τούτοις ὅτι ἅμα τὸ κινούμενον καὶ τὸ κινοῦν

evident in these that at the same time the moved and the moving

ἐστιν· ἐνυπάρχει γὰρ αὐτοῖς τὸ πρῶτον κινοῦν, ὥστ' οὐδέν

is: it exists for them the first moving, so that nothing

ἐστιν ἀναμεταξύ· ὅσα δ' ὑπ' ἄλλου κινεῖται, τετραχῶς ἀνάγκη

is between; as much but by another is moved, in four ways necessity

γίγνεσθαι· τέτταρα γὰρ εἴδη τῆς ὑπ' ἄλλου φορᾶς, ἕλξις, ὦσις,

to come to be: four for types of the by another motion, pulling, pushing,

ὄχησις, δίνησις. ἅπασαι γὰρ αἱ κατὰ τόπον κινήσεις

carrying, whirl. all for the in accordance with place movements

ἀνάγονται εἰς ταύτας· ἡ μὲν γὰρ ἔπωσις ὦσίς τίς ἐστιν,

lead back to these: the on the one hand for utterance is who is,

ὅταν τὸ ἀφ' αὑτοῦ κινοῦν ἐπακολουθοῦν ὠθῇ, ἡ δ' ἄπωσις, ὅταν

when the from itself moving following pushes, the and distance, when

μὴ ἐπακολουθῇ κινῆσαν, ἡ δὲ ῥῖψις, ὅταν σφο [243b] δροτέραν

not following to have moved, the but throw, when towards [243b] swifter

ποιήσῃ τὴν ἀφ' αὑτοῦ κίνησιν τῆς κατὰ φύσιν φορᾶς, καὶ

makes the from itself motion of the according to nature of movement, and

μέχρι τοσούτου φέρηται ἕως ἂν κρατῇ ἡ κίνησις. πάλιν

until so much is carried until whenever it prevails the motion. again

ἡ δίωσις καὶ σύνωσις ἄπωσις καὶ ἕλξις εἰσίν· ἡ μὲν

the pursuit and coalescence distance and attraction are: the on one hand

γὰρ δίωσις ἄπωσις (ἢ γὰρ ἀφ' αὑτοῦ ἢ ἀπ' ἄλλου ἐστὶν ἡ

for pursuit distance (either for from itself or from another it is the

ἄπωσις), ἡ δὲ σύνωσις ἕλξις (καὶ γὰρ πρὸς αὐτὸ καὶ πρὸς

distance), the but coalescence attraction (and for towards itself and towards

ἄλλο ἡ ἕλξις). ὥστε καὶ ὅσα τούτων εἴδη, οἷον σπάθησις

another the attraction). so and as many as of these types, such as tearing

καὶ κέρκισις· ἡ μὲν γὰρ σύνωσις, ἡ δὲ δίωσις. ὁμοίως δὲ καὶ

and cutting· the indeed for union, the but separation. similarly but also

αἱ ἄλλαι συγκρίσεις καὶ διακρίσεις—ἅπασαι γὰρ ἔσονται διώσεις

the other comparisons and distinctions-all for will be separations

ἢ συνώσεις—πλὴν ὅσαι ἐν γενέσει καὶ φθορᾷ εἰσιν. ἅμα δὲ

or unions-except as many as in generation and decay are. at once but

φανερὸν ὅτι οὐδ' ἔστιν ἄλλο τι γένος κινήσεως ἡ σύγκρισις
evident that not even there is another thing type of motion the comparison
καὶ διάκρισις· ἅπασαι γὰρ διανέμονται εἴς τινας τῶν εἰρημένων.
and distinction· all for are distributed into certain of the aforementioned.
ἔτι δ' ἡ μὲν εἰσπνοὴ ἕλξις, ἡ δ' ἐκπνοὴ ὦσις. ὁμοίως δὲ
further but the indeed inhalation pulling, the but exhalation pushing. similarly but
καὶ ἡ πτύσις, καὶ ὅσαι ἄλλαι διὰ τοῦ σώματος ἢ ἐκκριτικαὶ
also the expansion, and as many as other through the body or critical
ἢ ληπτικαὶ κινήσεις· αἱ μὲν γὰρ ἕλξεις εἰσίν, αἱ δ' ἀπώσεις.
or attractive movements the on one hand for pulls are, the but pushes.
δεῖ δὲ καὶ τὰς ἄλλας τὰς κατὰ τόπον ἀνάγειν· ἅπασαι
it is necessary and also the other the according to place to derive; all
γὰρ πίπτουσιν εἰς τέσσαρας ταύτας. τούτων δὲ πάλιν ἡ ὄχησις καὶ
for fall into four these. of these and again the motion and
ἡ δίνησις εἰς ἕλξιν καὶ ὦσιν. ἡ μὲν γὰρ ὄχησις κατὰ
the whirling into pulling and pushing. the on one hand for motion according to
τούτων τινὰ τῶν τριῶν τρόπων ἐστίν (τὸ μὲν γὰρ ὀχούμενον
these some of the three ways is (the on one hand for being moved
κινεῖται κατὰ συμβεβηκός, ὅτι ἐν κινουμένῳ ἐστὶν ἢ ἐπὶ
moves according to occurrence, because in moving it is or at
κινουμένου τινός, τὸ δ' ὀχοῦν ὀχεῖ ἢ ἑλκόμενον ἢ [244a]
of a moving something, the but moving conducts or being pulled or [244a]
ὠθούμενον ἢ δινούμενον, ὥστε κοινή ἐστιν ἁπασῶν τῶν τριῶν
being pushed or swirling, so that common is of all the three
ἡ ὄχησις)· ἡ δὲ δίνησις σύγκειται ἐξ ἕλξεώς τε καὶ ὤσεως·
the motion) the and whirl consists of pulling and and pushing.
ἀνάγκη γὰρ τὸ δινοῦν τὸ μὲν ἕλκειν τὸ δ' ὠθεῖν· τὸ μὲν
necessity for the whirling the on one hand pulling the but pushing. the on one hand

γὰρ ἀφ' αὑτοῦ τὸ δὲ πρὸς αὑτὸ ἄγει. ὥστ' εἰ τὸ ὠθοῦν καὶ
for from itself the and towards itself leads. therefore if the pushing and
τὸ ἕλκον ἅμα τῷ ὠθουμένῳ καὶ τῷ ἑλκομένῳ, φανερὸν
the pulling together with the pushed and with the pulled, evident
ὅτι τοῦ κατὰ τόπον κινουμένου καὶ κινοῦντος οὐδέν
that of the in accordance with place in motion and moving nothing
ἐστι μεταξύ. ἀλλὰ μὴν τοῦτο δῆλον καὶ ἐκ τῶν ὁρισμῶν· ὧσις
is in between. but indeed this clear and from the definitions. Motion
μὲν γάρ ἐστιν ἡ ἀφ' αὑτοῦ ἢ ἀπ' ἄλλου πρὸς ἄλλο κίνησις,
on one hand for is the from itself or from another towards another motion,
ἕλξις δὲ ἡ ἀπ' ἄλλου πρὸς αὑτὸ ἢ πρὸς ἄλλο, ὅταν θάττων
pulling but the from of another towards itself or towards another, whenever swifter
ἡ κίνησις ᾖ [τοῦ ἕλκοντος] τῆς χωριζούσης ἀπ' ἀλλήλων τὰ
the motion may be of the pulling] of the separating from each other the
συνεχῆ· οὕτω γὰρ συνεφέλκεται θάτερον. (τάχα δὲ δόξειεν
continuous· thus for it pulls together another. (perhaps and it might seem
ἂν εἶναί τις ἕλξις καὶ ἄλλως· τὸ γὰρ ξύλον ἕλκει τὸ πῦρ
would to be someone a pulling and otherwise· the for wood pulls the fire
οὐχ οὕτως. τὸ δ' οὐθὲν διαφέρει κινουμένου τοῦ ἕλκοντος ἢ
not in this way. the but nothing differs when moved of the pulling or
μένοντος ἕλκειν· ὁτὲ μὲν γὰρ ἕλκει οὗ ἔστιν, ὁτὲ
when stationary to pull· at one time indeed for it pulls where it is, at another
δὲ οὗ ἦν.) ἀδύνατον δὲ ἢ ἀφ' αὑτοῦ πρὸς ἄλλο ἢ ἀπ' ἄλλου
but where it was.) impossible and or from itself towards another or from another
πρὸς αὑτὸ κινεῖν [244b] μὴ ἁπτόμενον, ὥστε φανερὸν ὅτι τοῦ
towards itself to move [244b] not touching, so that it is evident that of the
κατὰ τόπον κινουμένου καὶ κινοῦντος οὐδέν ἐστι μεταξύ.
according to place of the moving and of the mover nothing is in between.

ἀλλὰ μὴν οὐδὲ τοῦ ἀλλοιουμένου καὶ τοῦ ἀλλοιοῦντος. τοῦτο δὲ

but indeed nor of the changing and of the altering. this but

δῆλον ἐξ ἐπαγωγῆς· ἐν ἅπασι γὰρ συμβαίνει ἅμα εἶναι

clear from induction; in all for it happens at the same time to be

τὸ ἔσχατον ἀλλοιοῦν καὶ τὸ πρῶτον ἀλλοιούμενον· <ὑπόκειται

the last altering and the first changing; it is subjected

γὰρ ἡμῖν τὸ τὰ ἀλλοιούμενα κατὰ τὰς παθητικὰς καλουμένας

for us the the changes according to the passive qualities

ποιότητας πάσχοντα ἀλλοιοῦσθαι>. ἅπαν γὰρ σῶμα σώματος

of quality; undergoing change.> all for body of the body

διαφέρει τοῖς αἰσθητοῖς ἢ πλείοσιν ἢ ἐλάττοσιν ἢ τῷ μᾶλλον καὶ

differs from the sensible or more or fewer or the more and

ἧττον τοῖς αὐτοῖς· ἀλλὰ μὴν καὶ ἀλλοιοῦται τὸ ἀλλοιούμενον ὑπὸ

less in the same; but indeed and it is altered the changing by

τῶν εἰρημένων. ταῦτα γάρ ἐστι πάθη τῆς ὑποκειμένης ποιότητος·

those mentioned. these for are passions of the underlying quality;

ἢ γὰρ θερμαινόμενον ἢ γλυκαινόμενον ἢ πυκνούμενον ἢ

or for that which is heated or that which is sweetened or that which is thickened or

ξηραινόμενον ἢ λευκαινόμενον ἀλλοιοῦσθαί φαμεν, ὁμοίως τό τε

drying or whitening to be altered we say, likewise the and

ἄψυχον καὶ τὸ ἔμψυχον λέγοντες, καὶ πάλιν τῶν ἐμψύχων τά τε

inanimate and the animate saying, and again of the animates the and

μὴ αἰσθητικὰ τῶν μερῶν καὶ αὐτὰς τὰς αἰσθήσεις. ἀλλοιοῦνται

not perceptible of the parts and themselves the senses. they are altered

γάρ πως καὶ αἱ αἰσθήσεις· ἡ γὰρ αἴσθησις ἡ κατ' ἐνέργειαν

for somehow and the senses; the for sense the through activity

κίνησίς ἐστι διὰ τοῦ σώματος, πασχούσης τι τῆς αἰσθήσεως.

motion is through the body, enduring something of the sense.

καθ' ὅσα μὲν οὖν τὸ ἄψυχον ἀλλοιοῦται, καὶ τὸ

according to as far as indeed therefore the inanimate is altered, and the

ἔμψυχον, καθ' ὅσα δὲ τὸ ἔμψυχον, οὐ κατὰ ταῦτα

animate, according to as far as but the animate, not according to these

πάντα τὸ ἄψυχον (οὐ γὰρ ἀλλοιοῦται κατὰ τὰς αἰσθήσεις)·

all the inanimate (not for it is altered according to the senses);

καὶ τὸ μὲν λανθάνει, τὸ δ' οὐ [245a] λανθάνει πάσχον.

and the on one hand it escapes, the that not [245a] escapes notice suffering.

οὐδὲν δὲ κωλύει καὶ τὸ ἔμψυχον λανθάνειν, ὅταν μὴ κατὰ

nothing and hinders and the living to escape notice, when not according to

τὰς αἰσθήσεις γίγνηται ἡ ἀλλοίωσις. εἴπερ οὖν ἀλλοιοῦται

the senses occurs the alteration. if indeed therefore is altered

τὸ ἀλλοιούμενον ὑπὸ τῶν αἰσθητῶν, ἐν ἅπασί γε τούτοις

the thing being altered by the sensibles, in all indeed these

φανερὸν ὅτι ἅμα ἐστὶ τὸ ἔσχατον ἀλλοιοῦν καὶ τὸ πρῶτον

evident that at the same time is the ultimate altering and the first

ἀλλοιούμενον· τῷ μὲν γὰρ συνεχὴς ὁ ἀήρ, τῷ δ' ἀέρι τὸ

being altered· to the indeed for continuous the air, to the and airy the

σῶμα. πάλιν δὲ τὸ μὲν χρῶμα τῷ φωτί, τὸ δὲ φῶς τῇ ὄψει.

body. again and the indeed color to the light, the and light by the sight.

τὸν αὐτὸν δὲ τρόπον καὶ ἡ ἀκοὴ καὶ ἡ ὄσφρησις· πρῶτον

the same and manner and the hearing and the smelling· first

γὰρ κινοῦν πρὸς τὸ κινούμενον ὁ ἀήρ. καὶ ἐπὶ τῆς γεύσεως

for to move towards the moving the air and upon the taste

ὁμοίως· ἅμα γὰρ τῇ γεύσει ὁ χυμός. ὡσαύτως δὲ καὶ

similarly at the same time for the taste the juice likewise but and

ἐπὶ τῶν ἀψύχων καὶ ἀναισθήτων. ὥστ' οὐδὲν ἔσται μεταξὺ τοῦ
upon the lifeless and insensate so that nothing will be in between the
ἀλλοιουμένου καὶ τοῦ ἀλλοιοῦντος.
being altered and the altering

οὐδὲ μὴν τοῦ αὐξανομένου τε καὶ αὔξοντος· αὐξάνει γὰρ τὸ
neither indeed of the increasing and and increasing. increases for the
πρῶτον αὔξον προσγιγνόμενον, ὥστε ἓν γίγνεσθαι τὸ ὅλον. καὶ
first increase coming to be, so that one becomes the whole. And
πάλιν φθίνει τὸ φθῖνον ἀπογιγνομένου τινὸς τῶν τοῦ φθίνοντος.
again decreases the dwindling coming to be of some of the the dwindling.
ἀνάγκη οὖν συνεχὲς εἶναι καὶ τὸ αὔξον καὶ τὸ φθῖνον, τῶν
Necessity therefore is continuous to be and the increasing and the dwindling, of the
δὲ συνεχῶν οὐδὲν μεταξύ.
but continuous nothing in between.

φανερὸν οὖν ὅτι τοῦ κινουμένου καὶ [245b] τοῦ κινοῦντος
It is evident therefore that of the moving and [245b] of the moving
πρώτου καὶ ἐσχάτου πρὸς τὸ κινούμενον οὐδέν ἐστιν ἀνὰ
first and last in relation to the moved nothing is in between
μέσον.
the middle.

Κεφάλαιο 3

Chapter 3

Ὅτι δὲ τὸ ἀλλοιούμενον ἅπαν ἀλλοιοῦται ὑπὸ τῶν αἰσθητῶν, καὶ
That but the undergoing alteration all is altered by the perceivable, and

ἐν μόνοις ὑπάρχει τούτοις ἀλλοίωσις ὅσα καθ' αὐτὰ λέγεται

in alone exists in these alteration as much as according to these is said

πάσχειν ὑπὸ τῶν αἰσθητῶν, ἐκ τῶνδε θεωρητέον. τῶν γὰρ

to suffer by the sensibles, out of these must be considered. of the for

ἄλλων μάλιστ' ἄν τις ὑπολάβοι ἔν τε τοῖς σχήμασι καὶ ταῖς

others especially if anyone were to suppose in both the shapes and the

μορφαῖς καὶ ἐν ταῖς ἕξεσι καὶ ταῖς τούτων λήψεσι καὶ ἀποβολαῖς

forms and in the habits and the of these perceptions and rejections

ἀλλοίωσιν ὑπάρχειν· ἐν οὐδετέροις δ' ἔστιν.

alterations to exist; in none other but it is.

τὸ μὲν γὰρ σχηματιζόμενον καὶ ῥυθμιζόμενον ὅταν ἐπιτελεσθῇ,

the on the one hand for being formed and being arranged when it is completed,

οὐ λέγομεν ἐκεῖνο ἐξ οὗ ἐστιν, οἷον τὸν ἀνδριάντα χαλκὸν ἢ

not we say that from which it is, such as the statue bronze or

τὴν πυραμίδα κηρὸν ἢ τὴν κλίνην ξύλον, ἀλλὰ παρωνυμιάζοντες

the pyramid wax or the bed wood, but calling by names

τὸ μὲν χαλκοῦν, τὸ δὲ κήρινον, τὸ δὲ

the on the one hand bronze, the on the other hand waxen, the on the other hand

ξύλινον. τὸ δὲ πεπονθὸς καὶ ἠλλοιωμένον προσαγορεύομεν·

wooden. the on the other hand having suffered and having been altered we name;

ὑγρὸν γὰρ καὶ θερμὸν καὶ σκληρὸν τὸν χαλκὸν λέγομεν καὶ τὸν

wet for and hot and hard the bronze we say and the

κηρόν

wax

(καὶ οὐ μόνον οὕτως, ἀλλὰ καὶ τὸ ὑγρὸν καὶ τὸ θερμὸν χαλκὸν

(and not only in this way, but also the wet and the hot bronze

λέγομεν), ὁμωνύμως τῷ πάθει προσαγορεύοντες τὴν [246a] ὕλην.

we say), homonymously to the suffering naming the [246a] matter.

ὥστ' εἰ κατὰ μὲν τὸ σχῆμα καὶ τὴν μορφὴν οὐ λέγεται

Therefore, if according to one hand the shape and the form not is said

τὸ γεγονὸς ἐν ᾧ ἐστι τὸ σχῆμα, κατὰ δὲ τὰ πάθη καὶ

the occurring in which is the shape, according to but the passions and

τὰς ἀλλοιώσεις λέγεται, φανερὸν ὅτι οὐκ ἂν εἶεν αἱ γενέσεις

the alterations is said, evident that not would be the births

ἀλλοιώσεις.

alterations.

ἔτι δὲ καὶ εἰπεῖν οὕτως ἄτοπον ἂν δόξειεν, ἠλλοιῶσθαι

yet but also to say thus would be strange would seem, to be altered

τὸν ἄνθρωπον ἢ τὴν οἰκίαν ἢ ἄλλο ὁτιοῦν τῶν γεγενημένων·

the human or the house or another anything of the generated;

ἀλλὰ γίγνεσθαι μὲν ἴσως ἕκαστον ἀναγκαῖον ἀλλοιουμένου

but to become indeed perhaps each necessary of something changing,

τινός, οἷον τῆς ὕλης πυκνουμένης ἢ μανουμένης ἢ θερμαινομένης

of something, such as the matter being condensed or being rarefied or being heated

ἢ ψυχομένης, οὐ μέντοι τὰ γιγνόμενά γε ἀλλοιοῦται, οὐδ' ἡ

or being cooled, not however the generated indeed are altered, nor the

γένεσις αὐτῶν ἀλλοίωσίς ἐστιν.

birth of them alteration is.

ἀλλὰ μὴν οὐδ' αἱ ἕξεις οὔθ' αἱ τοῦ σώματος οὔθ' αἱ τῆς ψυχῆς

but indeed nor the states nor the of the body nor the of the soul

ἀλλοιώσεις. αἱ μὲν γὰρ ἀρεταὶ αἱ δὲ κακίαι τῶν ἕξεων·
alterations. the on one hand for virtues the but vices of the states:
οὐκ ἔστι δὲ οὔτε ἡ ἀρετὴ οὔτε ἡ κακία ἀλλοίωσις, ἀλλ' ἡ
not is but neither the virtue nor the vice alteration, but the
μὲν ἀρετὴ τελείωσίς τις (ὅταν γὰρ λάβῃ τὴν αὑτοῦ
on one hand virtue perfection some (when for one receives the one's own
ἀρετήν, τότε λέγεται τέλειον ἕκαστον—τότε γὰρ ἔστι μάλιστα
virtue, then it is said perfect each-then for is especially
[τὸ] κατὰ φύσιν—ὥσπερ κύκλος τέλειος, ὅταν μάλιστα γένηται
[the] according to nature-as circle perfect, when especially it becomes
κύκλος καὶ ὅταν βέλτιστος), ἡ δὲ κακία φθορὰ τούτου καὶ
circle and when best), the but vice is destruction of this and
ἔκστασις· ὥσπερ οὖν οὐδὲ τὸ τῆς οἰκίας τελείωμα λέγομεν
ecstasy: just as therefore neither the of the household completion we call
ἀλλοίωσιν (ἄτοπον γὰρ εἰ ὁ θριγκὸς καὶ ὁ κέραμος ἀλλοίωσις,
alteration (strange for if the threshold and the clay alteration,
ἢ εἰ θριγκουμένη καὶ κεραμουμένη ἀλλοιοῦται ἀλλὰ μὴ τελειοῦται
or if being altered and being of clay is altered but not is perfected
ἡ οἰκία), τὸν αὐτὸν τρόπον καὶ ἐπὶ τῶν ἀρε [246b] τῶν καὶ
the house), the same way and upon the virtues [246b] the and
τῶν κακιῶν καὶ τῶν ἐχόντων ἢ λαμβανόντων· αἱ μὲν γὰρ
the vices and the of those having or of those receiving· the indeed for
τελειώσεις αἱ δὲ ἐκστάσεις εἰσίν, ὥστ' οὐκ ἀλλοιώσεις. ἔτι
perfections the but states of ecstasy are, so that not alterations. yet
δὲ καί φαμεν ἁπάσας εἶναι τὰς ἀρετὰς ἐν τῷ πρός τι πὼς
but also we say all to be the virtues in the towards something how
ἔχειν. τὰς μὲν γὰρ τοῦ σώματος, οἷον ὑγίειαν καὶ εὐεξίαν, ἐν
to hold. the indeed for of the body, such as health and well-being, in

κράσει καὶ συμμετρίᾳ θερμῶν καὶ ψυχρῶν τίθεμεν, ἢ αὐτῶν πρὸς

balance and symmetry of heat and of cold we place, or of these toward

αὐτὰ τῶν ἐντὸς ἢ πρὸς τὸ περιέχον· ὁμοίως δὲ καὶ τὸ κάλλος

themselves of the within or toward the encompassing; similarly and also the beauty

καὶ τὴν ἰσχὺν καὶ τὰς ἄλλας ἀρετὰς καὶ κακίας. ἑκάστη γάρ ἐστι

and the strength and the other virtues and vices. each for is

τῷ πρός τι πὼς ἔχειν, καὶ περὶ τὰ οἰκεῖα πάθη εὖ

to the toward something somehow to have, and concerning the own affections well

ἢ κακῶς διατίθησι τὸ ἔχον· οἰκεῖα δ' ὑφ' ὧν γίγνεσθαι καὶ

or badly disposes the having; own but under which come to be and

φθείρεσθαι πέφυκεν. ἐπεὶ οὖν τὰ πρός τι οὔτε αὐτά ἐστιν

to be destroyed has been. since therefore the toward something neither they are

ἀλλοιώσεις, οὔτε ἔστιν αὐτῶν ἀλλοίωσις οὐδὲ γένεσις οὐδ' ὅλως

alterations, nor is there of them alteration nor generation nor at all

μεταβολὴ οὐδεμία, φανερὸν ὅτι οὔθ' αἱ ἕξεις οὔθ' αἱ τῶν ἕξεων

change none, clear that neither the states nor the of the states

ἀποβολαὶ καὶ λήψεις ἀλλοιώσεις εἰσίν, ἀλλὰ γίγνεσθαι μὲν

losses and acquisitions alterations are, but to become on one hand

ἴσως αὐτὰς καὶ φθείρεσθαι ἀλλοιουμένων τινῶν ἀνάγκη,

perhaps them and to be destroyed of those being altered of certain necessity,

καθάπερ καὶ τὸ εἶδος καὶ τὴν μορφήν, οἷον θερμῶν καὶ ψυχρῶν ἢ

just as and the form and the shape, such as of warm and cold or

ξηρῶν καὶ ὑγρῶν, ἢ ἐν οἷς τυγχάνουσιν οὖσαι πρώτοις. περὶ

dry and wet, or in which they happen to be being first. concerning

ταῦτα γὰρ ἑκάστη λέγεται κακία καὶ ἀρετή, ὑφ' ὧν ἀλλοιοῦσθαι

these for each is said badness and virtue, through which to be altered

πέφυκε τὸ ἔχον· ἡ μὲν γὰρ ἀρετὴ ποιεῖ ἢ ἀπαθὲς

has been born the possessing; the on one hand for virtue produces or without passion

ἢ ὡδὶ παθητικόν, ἡ δὲ κακία παθητικὸν ἢ ἐναντίως

or in this way being passive, the on the other hand badness is passive or conversely

ἀπαθές.

without passion.

ὁμοίως [247a] δὲ καὶ ἐπὶ τῶν τῆς ψυχῆς ἕξεων· ἅπασαι γὰρ

similarly [247a] but and concerning of the the soul states. all for

καὶ αὗται τῷ πρός τι πὼς ἔχειν, καὶ αἱ μὲν ἀρεταὶ

and these to the regarding something how to have, and the indeed virtues

τελειώσεις, αἱ δὲ κακίαι ἐκστάσεις. ἔτι δὲ ἡ μὲν ἀρετὴ εὖ

completions, the but vices states. furthermore but the indeed virtue well

διατίθησι πρὸς τὰ οἰκεῖα πάθη, ἡ δὲ κακία κακῶς. ὥστ' οὐδ'

arranges towards the proper affections, the but vice poorly. so that not even

αὗται ἔσονται ἀλλοιώσεις· οὐδὲ δὴ αἱ ἀποβολαὶ καὶ αἱ λήψεις

these will be changes. nor indeed the losses and the gains

αὐτῶν. γίγνεσθαι δ' αὐτὰς ἀναγκαῖον ἀλλοιουμένου τοῦ αἰσθητικοῦ

of them. to become but them necessary when changing the sensory

μέρους. ἀλλοιωθήσεται δ' ὑπὸ τῶν αἰσθητῶν· ἅπασα γὰρ ἡ ἠθικὴ

part. it will be changed but by the senses. all for the ethical

ἀρετὴ περὶ ἡδονὰς καὶ λύπας τὰς σωματικάς, αὗται δὲ ἢ ἐν

virtue concerning pleasures and pains the bodily, these but or in

τῷ πράττειν ἢ ἐν τῷ μεμνῆσθαι ἢ ἐν τῷ ἐλπίζειν. αἱ μὲν οὖν

the doing or in the remembering or in the hoping. the indeed therefore

ἐν τῇ πράξει κατὰ τὴν αἴσθησίν εἰσιν, ὥσθ' ὑπ' αἰσθητοῦ

in the action according to the perception are, so that under sensible

τινὸς κινεῖσθαι, αἱ δ' ἐν τῇ μνήμῃ καὶ ἐν τῇ ἐλπίδι ἀπὸ ταύτης

some to be moved, the but in the memory and in the hope from this

εἰσίν· ἢ γὰρ οἷα ἔπαθον μεμνημένοι ἥδονται, ἢ ἐλπίζοντες

are: or for such as experienced remembering take pleasure, or hoping

οἷα μέλλουσιν. ὥστ' ἀνάγκη πᾶσαν τὴν τοιαύτην ἡδονὴν ὑπὸ τῶν

such as they will be. so that necessity every the such pleasure by the

αἰσθητῶν γίγνεσθαι. ἐπεὶ δ' ἡδονῆς καὶ λύπης ἐγγιγνομένης καὶ

sensibile to come about. since but of pleasure and pain coming into being and

ἡ κακία καὶ ἡ ἀρετὴ ἐγγίγνεται (περὶ ταύτας γάρ εἰσιν),

the badness and the virtue comes into being (concerning these for they are),

αἱ δ' ἡδοναὶ καὶ αἱ λῦπαι ἀλλοιώσεις τοῦ αἰσθητικοῦ, φανερὸν

the but pleasures and the pains alterations of the aesthetic, evident

ὅτι ἀλλοιουμένου τινὸς ἀνάγκη καὶ ταύτας ἀποβάλλειν καὶ

that when being altered of something necessity and these to cast off and

λαμβάνειν. ὥσθ' ἡ μὲν γένεσις αὐτῶν μετ' ἀλλοιώσεως, αὐταὶ

to take. Therefore, the indeed origin of them with alteration, they

δ' οὐκ εἰσὶν ἀλλοιώσεις.

but not are alterations.

[247b] ἀλλὰ μὴν οὐδ' αἱ τοῦ νοητικοῦ μέρους ἕξεις ἀλλοιώσεις,

[247b] but indeed nor the of the intellectual part you will have changes,

οὐδ' ἔστιν αὐτῶν γένεσις. πολὺ γὰρ μάλιστα τὸ ἐπιστῆμον ἐν τῷ

nor is of them generation. much for especially the scientific in the

πρός τι πὼς ἔχειν λέγομεν. ἔτι δὲ καὶ φανερὸν ὅτι οὐκ ἔστιν

context of some how to have we say. yet but also evident that not is

αὐτῶν γένεσις· τὸ γὰρ κατὰ δύναμιν ἐπιστῆμον οὐδὲν αὐτὸ

of them generation; the for according to ability scientific nothing it

κινηθὲν ἀλλὰ τῷ ἄλλο ὑπάρξαι γίγνεται ἐπιστῆμον. ὅταν γὰρ

having been moved but to the other to exist becomes scientific. when for

γένηται τὸ κατὰ μέρος, ἐπίσταταί πως τὰ καθόλου τῷ

it becomes the according to part, it knows somehow the universals in the

ἐν μέρει. πάλιν δὲ τῆς χρήσεως καὶ τῆς ἐνεργείας οὐκ ἔστι

in part. again but of the use and of the activity not is

γένεσις, εἰ μή τις καὶ τῆς ἀναβλέψεως καὶ τῆς ἁφῆς οἴεται

generation, if not someone and of the sight and of the touch thinks

γένεσιν εἶναι· τὸ γὰρ χρῆσθαι καὶ τὸ ἐνεργεῖν ὅμοιον τούτοις. ἡ

origin to be. the for to use and the acting similar to these. the

δ' ἐξ ἀρχῆς λῆψις τῆς ἐπιστήμης γένεσις οὐκ ἔστιν οὐδ'

but from the beginning assumption of the knowledge origin not is nor

ἀλλοίωσις· τῷ γὰρ ἠρεμῆσαι καὶ στῆναι τὴν διάνοιαν ἐπίστασθαι

alteration. to the for to be at rest and to stand the mind to understand

καὶ φρονεῖν λεγόμεθα, εἰς δὲ τὸ ἠρεμεῖν οὐκ ἔστι γένεσις·

and to think we are called, into but the being at rest not is origin.

ὅλως γὰρ οὐδεμιᾶς μεταβολῆς, καθάπερ εἴρηται πρότερον.

altogether for of no change, as has been said earlier.

ἔτι δ' ὥσπερ ὅταν ἐκ τοῦ μεθύειν ἢ καθεύδειν ἢ νοσεῖν

still but just as when out of the drunkenness or sleeping or being ill

εἰς τἀναντία μεταστῇ τις, οὔ φαμεν ἐπιστήμονα γεγονέναι

into opposites one changes, someone, not we say a sage has become

πάλιν (καίτοι ἀδύνατος ἦν τῇ ἐπιστήμῃ χρῆσθαι πρότερον),

again (and yet impossible was to the knowledge to use earlier),

οὕτως οὐδ' ὅταν ἐξ ἀρχῆς λαμβάνῃ τὴν ἕξιν· τῷ γὰρ

in this way nor when from the beginning takes the disposition; for indeed

καθίστασθαι τὴν ψυχὴν ἐκ τῆς φυσικῆς ταραχῆς φρόνιμόν τι

to be established the soul from the natural disturbance prudent something

γίγνεται καὶ ἐπιστῆμον. διὸ καὶ τὰ παιδία οὔτε μανθάνειν

occurs and knowledgeable. Therefore and the children neither to learn

δύνανται οὔτε κατὰ τὰς αἰσθήσεις [248a] ὁμοίως κρίνειν τοῖς

are able nor according to the senses [248a] similarly to judge the

πρεσβυτέροις· πολλὴ γὰρ ἡ ταραχὴ καὶ ἡ κίνησις. καθίσταται

elders; much for the disturbance and the motion. it is established

δὲ καὶ ἠρεμίζεται πρὸς ἔνια μὲν ὑπὸ τῆς φύσεως αὐτῆς,

but also is calmed towards some on the one hand by the nature itself,

πρὸς ἔνια δ' ὑπ' ἄλλων, ἐν ἀμφοτέροις δὲ ἀλλοιουμένων τινῶν

towards some but under others, in both and being changed some

τῶν ἐν τῷ σώματι, καθάπερ ἐπὶ τῆς χρήσεως καὶ τῆς ἐνεργείας,

of the in the body, just as in the usage and the action,

ὅταν νήφων γένηται καὶ ἐγερθῇ. φανερὸν οὖν ἐκ τῶν εἰρημένων

when sober becomes and awakens. evident therefore from of the things said

ὅτι τὸ ἀλλοιοῦσθαι καὶ ἡ ἀλλοίωσις ἔν τε τοῖς αἰσθητοῖς

that the becoming different and the alteration in and the sensible

γίγνεται καὶ ἐν τῷ αἰσθητικῷ μορίῳ τῆς ψυχῆς, ἐν ἄλλῳ δ'

occurs and in the sensitive part of the soul, in another but

οὐδενὶ πλὴν κατὰ συμβεβηκός.

to no one except according to what has happened.

Κεφάλαιο 4

Chapter 4

Ἀπορήσειε δ' ἄν τις πότερόν ἐστι κίνησις πᾶσα πάσῃ συμβλητὴ

Will he be puzzled and if anyone either is motion all every that which can be combi

ἢ οὔ. εἰ δή ἐστιν πᾶσα συμβλητή, καὶ ὁμοταχὲς τὸ ἐν

or not. if indeed it exists all that which can be combined, and is analogous the in

ἴσῳ χρόνῳ ἴσον κινούμενον, ἔσται περιφερής τις ἴση εὐθείᾳ,

equal time equal moving, it will be circular any equal, straight,

καὶ μείζων δὴ καὶ ἐλάττων. ἔτι ἀλλοίωσις καὶ φορά τις

and greater indeed and lesser. Furthermore, alteration and change any

ἴση, ὅταν ἐν ἴσῳ χρόνῳ τὸ μὲν ἀλλοιωθῇ τὸ δ' ἐνεχθῇ.

equal, when in equal time the on the one hand is altered the and is affected.

ἔσται ἄρα ἴσον πάθος μήκει. ἀλλ' ἀδύνατον. ἀλλ' ἆρα ὅταν

It will be therefore equal condition in length. But impossible. But will it when

ἐν ἴσῳ ἴσον κινηθῇ, τότε ἰσοταχές, ἴσον δ' οὐκ ἔστιν πάθος

in equal equal is moved, then it will be uniform, equal and not exists condition

μήκει, ὥστε οὐκ ἔστιν ἀλλοίωσις φορᾷ ἴση οὐδ' ἐλάττων, ὥστ'

in length, so that not exists alteration in form equal neither lesser, so that

οὐ πᾶσα συμβλητή;

not every being joined?

ἐπὶ δὲ τοῦ κύκλου καὶ τῆς εὐθείας πῶς συμβήσεται; ἄτοπόν τε

on and of the circle and of the straight line how will it happen? strange and

γὰρ εἰ μὴ ἔστιν κύκλῳ ὁμοίως τουτὶ κινεῖσθαι καὶ τουτὶ ἐπὶ τῆς

for if not it exists in a circle similarly this to move and this on the

εὐθείας, ἀλλ' εὐθὺς ἀνάγκη ἢ θᾶττον ἢ βραδύτερον, ὥσπερ ἂν εἰ

straight line, but straight necessity or faster or slower, just as if if

κάταντες, τὸ δ' ἄναντες· οὐδὲ διαφέρει οὐδὲν τῷ λόγῳ,

they are descending, the but opposite; nor does it differ anything to the argument,

εἴ τίς φησιν ἀνάγκην εἶναι θᾶττον εὐθὺς ἢ βραδύτερον κινεῖσθαι·

if anyone says necessity to be faster straight or slower to move;

ἔσται γὰρ μείζων καὶ ἐλάττων ἡ περιφερὴς τῆς εὐθείας,

it will be for greater and lesser the circumference of the straight line,

ὥστε καὶ ἴση. εἰ γὰρ ἐν τῷ A χρόνῳ [248b] τὴν B διελήλυθε

so that also it is equal. if for in the A time [248b] the B has passed

τὸ δὲ τὴν Γ, μείζων ἂν εἴη ἡ Β τῆς Γ· οὕτω γὰρ τὸ

the but the G, greater would let it be the B of the G. thus for the

θᾶττον ἐλέγετο. οὐκοῦν καὶ εἰ ἐν ἐλάττονι ἴσον, θᾶττον· ὥστ'

swifter was said. therefore and if in lesser equal, swifter. so that

ἔσται τι μέρος τοῦ Α ἐν ᾧ τὸ Β τοῦ κύκλου τὸ ἴσον δίεισι

it will be some part of the A in which the B of the circle the equal passes

καὶ τὸ Γ ἐν ὅλῳ τῷ Α [τὴν Γ]. ἀλλὰ μὴν εἰ ἔστιν συμβλητά,

and the G in whole the A the G]. but indeed if it exists combinable,

συμβαίνει τὸ ἄρτι ῥηθέν, ἴσην εὐθεῖαν εἶναι κύκλῳ. ἀλλ' οὐ

it happens the now stated, to be equal straight to be to a circle. but not

συμβλητά· οὐδ' ἄρα αἱ κινήσεις, ἀλλ' ὅσα μὴ συνώνυμα,

combinable. nor therefore the motions, but as much as not synonymous,

πάντ' ἀσύμβλητα.

all incompatible.

οἷον διὰ τί οὐ συμβλητὸν πότερον ὀξύτερον τὸ γραφεῖον ἢ ὁ

such as for what not comparable either sharper the writing or the

οἶνος ἢ ἡ νήτη; ὅτι ὁμώνυμα, οὐ συμβλητά· ἀλλ' ἡ

wine or the grape? because they are homonymous, not comparable. but the

νήτη τῇ παρανήτῃ συμβλητή, ὅτι τὸ αὐτὸ σημαίνει τὸ ὀξὺ ἐπ'

grape to the vine is comparable, because the same signifies the sharp in

ἀμφοῖν. ἆρ' οὖν οὐ ταὐτὸν τὸ ταχὺ ἐνταῦθα κἀκεῖ, πολὺ δ' ἔτι

both. is it then not the same the swift here and there, much but still

ἧττον ἐν ἀλλοιώσει καὶ φορᾷ;

less in alteration and form?

ἢ πρῶτον μὲν τοῦτο οὐκ ἀληθές, ὡς εἰ μὴ ὁμώνυμα συμβλητά; τὸ
Or first indeed this not true, as if not homonyms applicable? The
γὰρ πολὺ τὸ αὐτὸ σημαίνει ἐν ὕδατι καὶ ἀέρι, καὶ οὐ συμβλητά.
for much the same signifies in water and air, and not applicable.
εἰ δὲ μή, τό γε διπλάσιον ταὐτό (δύο γὰρ πρὸς ἕν), καὶ οὐ
If but not, that indeed double same (two for towards one), and not
συμβλητά. ἢ καὶ ἐπὶ τούτων ὁ αὐτὸς λόγος; καὶ γὰρ τὸ πολὺ
applicable. Or also in these the same reasoning? And for the much
ὁμώνυμον. ἀλλ' ἐνίων καὶ οἱ λόγοι ὁμώνυμοι, οἷον εἰ λέγοι
homonymous. But of some and the reasons are homonymous, such as if someone says
τις ὅτι τὸ πολὺ τὸ τοσοῦτον καὶ ἔτι, ἄλλο τὸ τοσοῦτον· καὶ
anyone that the much the so much and more, another the so much; and
τὸ ἴσον ὁμώνυμον, καὶ τὸ ἓν δέ, εἰ ἔτυχεν, εὐθὺς ὁμώνυμον.
the equal homonymous, and the one but, if it happens, immediately homogeneous.
εἰ δὲ τοῦτο, καὶ τὰ δύο, ἐπεὶ διὰ τί τὰ μὲν συμβλητὰ τὰ
if but this, and the two, since through why the on one hand are comparable the
δ' οὔ, εἴπερ ἦν μία φύσις; ἢ ὅτι ἐν ἄλλῳ πρώτῳ
on the other hand no, if indeed it was one nature? or that in another first
δεκτικῷ; ὁ μὲν οὖν ἵππος καὶ ὁ κύων συμβλητά, πότερον
receptive? the on one hand therefore horse and the dog are comparable, whether
λευκότερον (ἐν ᾧ γὰρ πρώτῳ, τὸ αὐτό, ἡ ἐπιφάνεια), καὶ
whiter (in which for first, the same, the appearance), and
κατὰ μέγεθος ὡσαύτως· ὕδωρ δὲ καὶ φωνὴ οὔ· ἐν ἄλλῳ
according to size likewise. Water but and voice no; in another
γάρ. ἢ δῆλον ὅτι ἔσται οὕτω γε πάντα ἓν ποιεῖν, ἐν ἄλλῳ
for. Or it is clear that it will be thus indeed all one to make, in another
[249a] δὲ ἕκαστον φάσκειν εἶναι, καὶ ἔσται ταὐτὸ <τὸ> ἴσον καὶ
[249a] but each says to be, and will be the same <the> equal and

γλυκὺ καὶ λευκόν, ἀλλ' ἄλλο ἐν ἄλλῳ; ἔτι δεκτικὸν οὐ τὸ τυχὸν

sweet and white, but another in another; still acceptable not the possible

<τοῦ τυχόντος> ἐστίν, ἀλλ' ἓν ἑνὸς τὸ πρῶτον.

of the happening> is, but one of one the first.

ἀλλ' ἆρα οὐ μόνον δεῖ τὰ συμβλητὰ μὴ ὁμώνυμα εἶναι ἀλλὰ

but is it not only it is necessary the composite not homonymous to be but

καὶ μὴ ἔχειν διαφοράν, μήτε ὃ μήτε ἐν ᾧ; λέγω δὲ οἷον

and not to have a difference, nor that nor in which? I say but such as

χρῶμα ἔχει διαί ρεσιν· τοιγαροῦν οὐ συμβλητὸν κατὰ

color has separation of resin. therefore not composite according to

τοῦτο (οἷον πότερον κεχρωμάτισται μᾶλλον, μὴ κατὰ τὶ

this (such as whether it has been colored more, not according to what

χρῶμα, ἀλλ' ᾗ χρῶμα), ἀλλὰ κατὰ τὸ λευκόν.

color, but by which color), but according to the white.

οὕτω καὶ περὶ κίνησιν ὁμοταχὲς τῷ ἐν ἴσῳ χρόνῳ κινεῖσθαι

thus and concerning movement uniform to the in equal time to be moved

ἴσον τοσονδί· εἰ δὴ τοῦ μήκους ἐν τῳδὶ τὸ μὲν

equal to such an extent if indeed of the length in this the on one hand

ἠλλοιώθη τὸ δ' ἠνέχθη, ἴση ἄρα αὕτη ἡ ἀλλοίωσις καὶ

was altered the but was carried equal therefore this the alteration and

ὁμοταχὴς τῇ φορᾷ; ἀλλ' ἄτοπον. αἴτιον δ' ὅτι ἡ κίνησις ἔχει

uniform to the motion? but strange. cause but that the movement has

εἴδη, ὥστ' εἰ τὰ ἐν ἴσῳ χρόνῳ ἐνεχθέντα ἴσον μῆκος ἰσοταχῇ

types, so that if the in equal time conveyed equal length uniformly

ἔσται, ἴση ἡ εὐθεῖα καὶ ἡ περιφερής. πότερον οὖν αἴτιον, ὅτι ἡ

will be, equal the straight and the circular. whether then cause, that the

φορὰ γένος ἢ ὅτι ἡ γραμμὴ γένος; ὁ μὲν γὰρ χρόνος ὁ

motion is a kind or that the line is a kind? the on one hand for time the

αὐτός, ἂν δὲ τῷ εἴδει ᾖ ἄλλα, καὶ ἐκεῖνα εἴδει διαφέρει. καὶ

same, if but to the form it is other, and those forms differs. And

γὰρ ἡ φορὰ εἴδη ἔχει, ἂν ἐκεῖνο ἔχῃ εἴδη ἐφ' οὗ κινεῖται

indeed the movement forms has, if that one should have forms on which it moves

(ὁτὲ δὲ ἐὰν ᾧ, οἷον εἰ πόδες, βάδισις, εἰ δὲ πτέρυγες,

(whenever but if to which, as if feet, walking, if but wings,

πτῆσις. ἢ οὔ, ἀλλὰ τοῖς σχήμασιν ἡ φορὰ ἄλλη;). ὥστε

flying. Or not, but to the shapes the movement is different?). Therefore

τὰ ἐν ἴσῳ ταὐτὸ μέγεθος κινούμενα ἰσοταχῆ, τὸ αὐτὸ δὲ καὶ

the in equality the same size moving at equal speed, the same and also

ἀδιάφορον εἴδει καὶ κινήσει ἀδιάφορον· ὥστε τοῦτο σκεπτέον,

indifferent forms and motion indifferent. Therefore this must be considered,

τίς διαφορὰ κινήσεως. καὶ σημαίνει ὁ λόγος οὗτος ὅτι τὸ γένος

what difference of motion. And signifies this statement that that the kind

οὐχ ἕν τι, ἀλλὰ παρὰ τοῦτο λανθάνει πολλά, εἰσίν τε τῶν

not one thing, but beside this escapes many, they are and of the

ὁμωνυμιῶν αἱ μὲν πολὺ ἀπέχουσαι, αἱ δὲ ἔχουσαί

homonyms the on one hand much differing, the on the other hand having

τινα ὁμοιότητα, αἱ δ' ἐγγὺς ἢ γένει ἢ ἀναλογίᾳ, διὸ οὐ

some similarity, the and close or by kind or by analogy, therefore not

δοκοῦσιν ὁμωνυμίαι εἶναι οὖσαι.

they seem homonyms to be existing.

πότε οὖν ἕτερον τὸ εἶδος, ἐὰν ταὐτὸ ἐν ἄλλῳ, ἢ ἂν ἄλλο

when therefore another the kind, if the same in another, or if something else

ἐν ἄλλῳ; καὶ τίς ὅρος; ἢ τῷ κρινοῦμεν ὅτι ταὐτὸν τὸ

in another? and what definition? or by which we judge that the same the

λευκὸν καὶ τὸ γλυκὺ ἢ ἄλλο—ὅτι ἐν ἄλλῳ φαίνεται ἕτερον,

white and the sweet or something else-that in another appears as different,

ἢ ὅτι ὅλως οὐ ταὐτό; περὶ δὲ δὴ ἀλλοιώσεως, πῶς ἔσται

or that entirely not the same? about but indeed of alteration, how could it be

ἰσοταχὴς ἑτέρᾳ ἑτέρα; εἰ δή ἐστι τὸ ὑγιάζεσθαι ἀλλοιοῦσθαι,

isochronous to another? different? if indeed it is the being healthy to be altered,

ἔστι τὸν μὲν ταχὺ τὸν δὲ βραδέως ἰαθῆναι, καὶ ἅμα

it is the one quickly the the slowly to be healed, and at the same time

τινάς, ὥστ' [249b] ἔσται ἀλλοίωσις ἰσοταχής· ἐν ἴσῳ γὰρ

some, such that [249b] it will be alteration of isochronous; in equal for

χρόνῳ ἠλλοιώθη. ἀλλὰ τί ἠλλοιώθη; τὸ γὰρ ἴσον οὐκ ἔστιν

time was altered. But what was altered? the for equal is not it is

ἐνταῦθα λεγόμενον, ἀλλ' ὡς ἐν τῷ ποσῷ ἰσότης, ἐνταῦθα ὁμοιότης.

here called, but as in the quantity equality, here similarity.

ἀλλ' ἔστω ἰσοταχὲς τὸ ἐν ἴσῳ χρόνῳ τὸ αὐτὸ μεταβάλλον.

but let it be equal in speed the in equal time the same changing.

πότερον οὖν ἐν ᾧ τὸ πάθος ἢ τὸ πάθος δεῖ συμβάλλειν;

which therefore in which the condition or the condition must be contributed?

ἐνταῦθα μὲν δὴ ὅτι ὑγίεια ἡ αὐτή, ἔστιν λαβεῖν ὅτι οὔτε

here indeed then that health the same, it is to take that neither

μᾶλλον οὔτε ἧττον ἀλλ' ὁμοίως ὑπάρχει. ἐὰν δὲ τὸ πάθος ἄλλο

more nor less but similarly exists. if but the condition another

ᾖ, οἷον ἀλλοιοῦται τὸ λευκαινόμενον καὶ τὸ ὑγιαζόμενον, τούτοις

be, such as is altered the becoming white and the becoming healthy, to these

οὐδὲν τὸ αὐτὸ οὐδ' ἴσον οὐδ' ὅμοιον, ᾗ ἤδη ταῦτα εἴδη

nothing the same nor equal nor similar, to which already these kinds

ποιεῖ ἀλλοιώσεως, καὶ οὐκ ἔστι μία ὥσπερ οὐδ' αἱ φοραί. ὥστε

makes of alteration, and not it is one even as not even the instances. so that

ληπτέον πόσα εἴδη ἀλλοιώσεως καὶ πόσα φορᾶς. εἰ μὲν

it must be taken how many types of alteration and how many instances. if on one hand

οὖν τὰ κινούμενα εἴδει διαφέρει, ὧν εἰσὶν αἱ κινήσεις

therefore the moving types differ, of which there are the movements

καθ' αὑτὰ καὶ μὴ κατὰ συμβεβηκός, καὶ αἱ κινήσεις

according to themselves and not according to coincidental, and the movements

εἴδει διοίσουσιν· εἰ δὲ γένει, γένει, εἰ δ' ἀριθμῷ, ἀριθμῷ. ἀλλὰ

types will differ; if but by kind, by kind, if but by number, by number. however

δὴ πότερον εἰς τὸ πάθος δεῖ βλέψαι, ἐὰν ᾖ τὸ αὐτὸ ἢ

indeed which into the affection must be considered, if there be the same or

ὅμοιον, εἰ ἰσοταχεῖς αἱ ἀλλοιώσεις, ἢ εἰς τὸ ἀλλοιούμενον, οἷον

similar, if equidistant the alterations, or into the altered, such as

εἰ τοῦ μὲν τοσονδὶ λελεύκανται τοῦ δὲ τοσονδί; ἢ εἰς ἄμφω,

if of on one hand so much are whitened of the but so much? or into both,

καὶ ἡ αὐτὴ μὲν ἢ ἄλλη τῷ πάθει, εἰ τὸ αὐτὸ <ἢ μὴ τὸ>

and the same on one hand or another to the suffering, if the same <or not the>

αὐτό, ἴση δ' ἢ ἄνισος, εἰ ἐκεῖνο <ἴσον ἢ> ἄνισον;

same, equal but or unequal, if that <equal or> unequal;

καὶ ἐπὶ γενέσεως δὲ καὶ φθορᾶς τὸ αὐτὸ σκεπτέον. πῶς

and upon generation but and decay the same must be considered. how

ἰσοταχὴς ἡ γένεσις; εἰ ἐν ἴσῳ χρόνῳ τὸ αὐτὸ καὶ ἄτομον, οἷον

equal the generation? if in equal time the same and individual, such as

ἄνθρωπος ἀλλὰ μὴ ζῷον· θάττων δ', εἰ ἐν ἴσῳ ἕτερον (οὐ

human but not animal. swifter however, if in equal other (not

γὰρ ἔχομέν τινα δύο ἐν οἷς ἡ ἑτερότης ὡς ἡ ἀνομοιότης), ἤ, εἰ

for we possess some two in which the otherness as the inequality), or, if

ἔστιν ἀριθμὸς ἡ οὐσία, πλείων καὶ ἐλάττων ἀριθμὸς ὁμοειδής·

there is a number the essence, greater and lesser number of the same kind.

ἀλλ' ἀνώνυμον τὸ κοινόν, καὶ τὸ ἑκάτερον [ποιόν· τὸ μὲν

but nameless the common, and the each [quality; the on one hand

ποιόν,] ὥσπερ τὸ πλεῖον πάθος ἢ τὸ ὑπερέχον μᾶλλον, τὸ δὲ

quality,] just as the greater phenomenon or the exceeding more, the but

ποσὸν μεῖζον.

quantity greater.

Κεφάλαιο 5

Chapter 5

εἰ δὲ τὰ <κινοῦντα> δύο, καὶ ἑκάτερον τῶνδε ἑκάτερον κινεῖ τὸ

if but the moving two, and each of these each moves the

τοσόνδε ἐν τοσῷδε, καὶ συντιθέμεναι αἱ δυνάμεις τὸ σύνθετον ἐκ

this amount in this much, and being composed the forces the composite from

τῶν βαρῶν τὸ ἴσον κινήσουσιν μῆκος καὶ ἐν ἴσῳ χρόνῳ· ἀνάλογον

the heavy the equal will move length and in equal time. proportional

γάρ.

indeed.

ἆρ’ οὖν οὕτω καὶ ἐπ’ ἀλλοιώσεως καὶ ἐπ’ αὐξήσεως; τὶ μὲν
Is therefore thus and upon alteration and upon increase? what indeed
γὰρ τὸ αὔξον, τὶ δὲ τὸ αὐξανόμενον, ἐν ποσῷ δὲ χρόνῳ καὶ
for the increaser, what but the being increased, in amount and time and
ποσὸν τὸ μὲν αὔξει τὸ δὲ αὐξάνεται. καὶ τὸ ἀλλοιοῦν καὶ
amount the indeed increases the while is being increased. And the othering and
τὸ ἀλλοιούμενον ὡσαύτως—τὶ καὶ ποσὸν [250b] κατὰ τὸ μᾶλλον
the being altered likewise-what and amount [250b] according to the more
καὶ ἧττον ἠλλοίωται, καὶ ἐν ποσῷ χρόνῳ, ἐν διπλασίῳ διπλάσιον,
and less is altered, and in how much time, in double double,
καὶ τὸ διπλάσιον ἐν διπλασίῳ· τὸ δ’ ἥμισυ ἐν ἡμίσει χρόνῳ (ἢ ἐν
and the double in double. the but half in half time (or in
ἡμίσει ἥμισυ), ἢ ἐν ἴσῳ διπλάσιον.
half half), or in equal double.

εἰ δὲ τὸ ἀλλοιοῦν ἢ αὔξον τὸ τοσόνδε ἐν τῷ τοσῷδε αὔξει ἢ
if but the changing or increasing the this much in the this much increases or
ἀλλοιοῖ, οὐκ ἀνάγκη καὶ τὸ ἥμισυ ἐν ἡμίσει καὶ ἐν ἡμίσει ἥμισυ,
changes, not necessary and the half in half and in half half,
ἀλλ’ οὐδέν, εἰ ἔτυχεν, ἀλλοιώσει ἢ αὐξήσει, ὥσπερ καὶ ἐπὶ τοῦ
but nothing, if it happens, alters or increases, just as also in the
βάρους.
weight.